

High-Level Design (HLD)

RAG-Based Customer Support Assistant with LangGraph & HITL

1. System Overview

This system is a Retrieval-Augmented Generation (RAG) based assistant designed to answer customer queries using a PDF knowledge base. It retrieves relevant content using embeddings and generates accurate responses using an LLM. Queries with low confidence are escalated to a Human-in-the-Loop (HITL) system.

2. Problem Statement

Traditional chatbots often provide incorrect answers due to lack of context. This system improves accuracy by retrieving real data from documents before generating responses.

3. System Scope

The system processes a PDF knowledge base and answers user queries. It supports workflow control using LangGraph and includes escalation to human agents for complex queries.

4. System Architecture Overview

User Query → Input Node → Retriever → Vector Database → LLM Processing → Decision Layer → Output or HITL Escalation.

5. Core Components

• Document Loader: Loads PDF data • Chunking Module: Splits text into smaller parts • Embedding Model: Converts text into vectors • Vector Database: Stores embeddings • Retriever: Finds relevant chunks • LLM: Generates answers • Workflow Engine: Controls execution flow • Routing Logic: Decides response or escalation • HITL Module: Handles complex queries

6. Data Flow

PDF → Chunking → Embeddings → Storage → Query → Retrieval → LLM → Response → (Optional HITL escalation)

7. Technology Stack

Python, LangChain, ChromaDB, Sentence Transformers, LangGraph

8. Scalability Considerations

Supports larger documents, increased user queries, and can be extended with cloud-based vector databases and optimized retrieval techniques.